Project 3: Convolutional Neural Networks and Image Classification

Settlers of Catan
December 12, 2025

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Introduction

Photography varies from person to person because of angles, style, objects, and compositional techniques. In this project, we explored whether a deep learning model could learn to identify the difference between two photographers Alex and Kelly based on the visual content of their images. We built a classification model that predicts which photographer took images on unseen test images and wanted to identify which visual features and composition styles correlate with each photographer, ultimately to understand what the model learned.

We were given over 400 labeled training images, 74 images for a dev set, and two holdout sets for evaluation. The training data had about 200 images from Alex and 200 images from Kelly. Holdout Set 1 had 20 unlabeled images for initial model evaluation, and Holdout Set 2 had 46 unlabeled images for class ensemble predictions. All images were in common formats like PNG and JPG with varying resolutions and ratios. We couldn't use EXIF data or file properties.

Data Preparation

All images were preprocessed to ensure consistent input formatting and compatibility with the pretrained DINOv2 vision transformer model. For each image we first loaded and converted it to RGB format to ensure a consistent three-channel input. Then we resized and center-cropped all images to a fixed resolution of 224x224 pixels. This step standardizes image dimensions across the dataset, matching the input requirements of the pretrained DINOv2 vision transformer model.

After resizing, images were converted into tensors and their pixel values were scaled from the range [0, 255] to [0,1]. We then normalized each color channel using a mean of 0.5 and a standard deviation of 0.5 to rescale the pixel values to approximately [-1, 1] to align with the distribution DINOv2 was originally trained on and ensure more stable feature extraction.

We also labeled training images with 10 binary features from two different categories. Our content features included whether the image was taken in Europe, was a close-up shot, featured young children, featured Kelly's husband, or featured board games. Our composition features included one-point perspective, framing techniques, vertical lines, vast distribution of color, and flat perspective (*see Appendix A*). We chose these features by looking through many images from each photographer, and because we identified both subject matter and compositional technique differences that seem to strongly appear in either Alex or Kelly.

Feature	Description
Vertical Lines Emphasis	Image contains strong vertical structures such as buildings, columns, or trees that dominate the composition.
One Point Perspective	The scene exhibits clear depth with lines converging toward a single vanishing point.
Flat Perspective	Minimal depth or perspective cues; scene appears front-facing or compressed.
Framing	The subject is visually framed by surrounding elements such as windows, arches, foliage, or foreground objects.
Vast Color Distribution	The image contains a wide range of vivid or saturated colors across the scene.
European Look	The scene features architecture, streets, or landmarks commonly associated with European locations.
Close-up of People	The image focuses closely on one or more people, with faces or upper bodies occupying a large portion of the frame.
Young Children	The image prominently features one or more children.
Kelly’s husband	The image includes Kelly’s husband.
Board Games	The image prominently features board games.

Table 1. Hand Picked Feature Descriptions

We split our training data by using 326 images which is 80% of the data to train the SVM and held out 74 images which is 20% as a dev set to test performance. It was a random split and had about equal images of Alex and Kelly photos in each set. Finally, we tested on Holdout Set 1 with 20 images and Holdout Set 2 with 46 images for evaluation.

Methods: Phase 1

The classification approach consisted of a pipeline including generating embeddings through DINOv2 and classifying them using a Support Vector Machine. DINOv2 is a self-supervised vision transformer model developed by Facebook Research that has been trained on millions of images to learn general visual representations. We specifically used DINOv2-ViT-S/14. These embeddings detect patterns like compositions, colors, textures, and objects. Then, we trained a Support Vector Machine to classify images using these embeddings. We chose a Support Vector Machine (SVM) classifier due to its effectiveness in high-dimensional feature spaces and its ability to model non-linear decision boundaries. DINOv2 embeddings are high-dimensional representations, making SVMs a suitable choice for separating stylistic differences between Alex’s and Kelly’s photographs. Each image was loaded, preprocessed according to DINOv2's requirements, and passed through the pre-trained transformer to extract its embedding. These embeddings were then used as input features for the SVM classifier. Model training and evaluation were implemented using scikit-learn, with hyperparameters selected through development set validation.

Methods: Phase 2

To better understand which visual characteristics distinguish Alex's photographs from Kelly's, we manually annotated the training images with ten binary features described in the Data Preparation section. We then trained a Random Forest classifier on these manually labeled features to estimate feature importance. This analysis was used purely for interpretability rather than as a final classification model.

In addition, we computed the percentage of images from each photographer that exhibited each feature, allowing us to compare stylistic tendencies quantitatively. These comparisons revealed clear stylistic differences between the two photographers. For example, approximately 55% of Alex's photos were taken in Europe, compared to only 5% of Kelly's. In contrast, about 55% of Kelly's images featured a wide distribution of vivid colors, while only around 22% of Alex's images exhibited this characteristic. These patterns align with the feature importance results and suggest that the DINOv2 embeddings may be capturing similar compositional and content-based cues.

After ground truth labels were released for Holdout Set 1, we conducted an exploratory experiment treating it as additional labeled data to assess whether increasing the amount of training data would improve generalization. This experiment did not result in improved performance on Holdout Set 2, suggesting that model performance is limited by feature representation rather than training set size.

Results: Development Set Performance

We evaluated our model with Accuracy, Precision, Recall, and F1 Score. We optimised on accuracy and used precision and recall to give us insights on the model's bias toward one class over another. For the holdout sets, we measured confidence scores from the SVM's probability estimates that give how sure the model was about each prediction. These confidence scores ranged from 0 to 1, where values close to 0 indicated high confidence for Alex and values close to 1 indicated high confidence for Kelly.

	Alex	**Kelly**
Precision	0.739	0.821
Recall	0.872	0.657
F1 Score	0.800	0.730
Accuracy	**0.770**	

Table 2. SVM Model Metrics

Table 2 summarizes the performance of the SVM classifier on the development set. The model achieved an overall accuracy of 77% across 74 images, which indicates that it was able to capture some meaningful stylistic differences between Alex and Kelly's photographs. Performance however, slightly differs by class. The model shows higher recall for Alex (0.872), but slightly lower precision (0.739), meaning the model is able to correctly identify most of Alex's photographs, but some Kelly images are misclassified as Alex. In contrast, the model achieves higher precision (0.821) for Kelly, but lower recall (0.657) which means that while predictions for Kelly photos are usually correct, a larger portion of Kelly's photos are incorrectly classified as Alex. This asymmetry suggests that Alex's images are more visually consistent and easier for the model to detect while Kelly's images exhibit greater visual diversity. Overall, the F1 scores (0.800 for Alex and 0.730 for Kelly) reveal that the classifier performs reasonably well across both classes.

Predicted / True	**Alex**	**Kelly**
Alex	**34**	5
Kelly	12	**23**

Table 3. SVM Dev Set Confusion Matrix

Table 3 provides a better picture of how misclassifications occur in the development set. The majority of errors arise from Kelly images being predicted as Alex

(12), while the reverse error is less frequent (5), which reinforces the class-level asymmetry observed in Table 2. This suggests that when the model is uncertain, it more often predicts Alex, indicating that Alex's images are represented more consistently in the embedding space.

Results: Holdout Set Performance

To assess generalization beyond the development set, we evaluated the trained model on unseen holdout data. Holdout Set 1 was initially used for evaluation in Phase 1 and is presented here primarily for error analysis. Holdout Set 2 was reserved exclusively for final performance evaluation.

On Holdout Set 1, the model exhibited a similar error pattern to the development set. Out of 20 images, two Alex images were misclassified as Kelly, while six Kelly images were misclassified as Alex (*see Appendix B*). This continued bias toward predicting Alex further supports the hypothesis that Alex's images occupy a more compact region in the embedding space, whereas Kelly's images exhibit greater stylistic variability.

True \ Predicted	Alex	Kelly
Alex	**15**	8
Kelly	12	**11**

Table 4. SVM Holdout Set 2 Confusion Matrix

	Alex	Kelly
Precision	0.56	0.58
Recall	0.65	0.48
F1 Score	0.6	0.52
Accuracy	**0.565**	

Table 5. SVM Holdout Set 2 Model Metrics

Table 4 presents the confusion matrix for Holdout Set 2, which serves as the final evaluation dataset. Performance on Holdout Set 2 was lower than on the development set, reflecting the increased difficulty of generalizing stylistic classification to fully unseen images. As shown in Table 5, overall accuracy decreased to 56.5%, with higher recall for Alex (0.65) than for Kelly (0.48). This persistent class-level asymmetry indicates a continued bias toward predicting Alex, consistent with earlier results. These findings support the hypothesis that Alex's images occupy a more compact region in the embedding space, while Kelly's images exhibit greater stylistic variability.

After ground truth labels were released for Holdout Set 1, we conducted an exploratory experiment treating it as additional labeled training data to assess whether increasing the training set size would improve generalization. This experiment did not result in improved performance on Holdout Set 2, suggesting that model performance is limited by feature representation rather than training set size (*see Appendix C*).

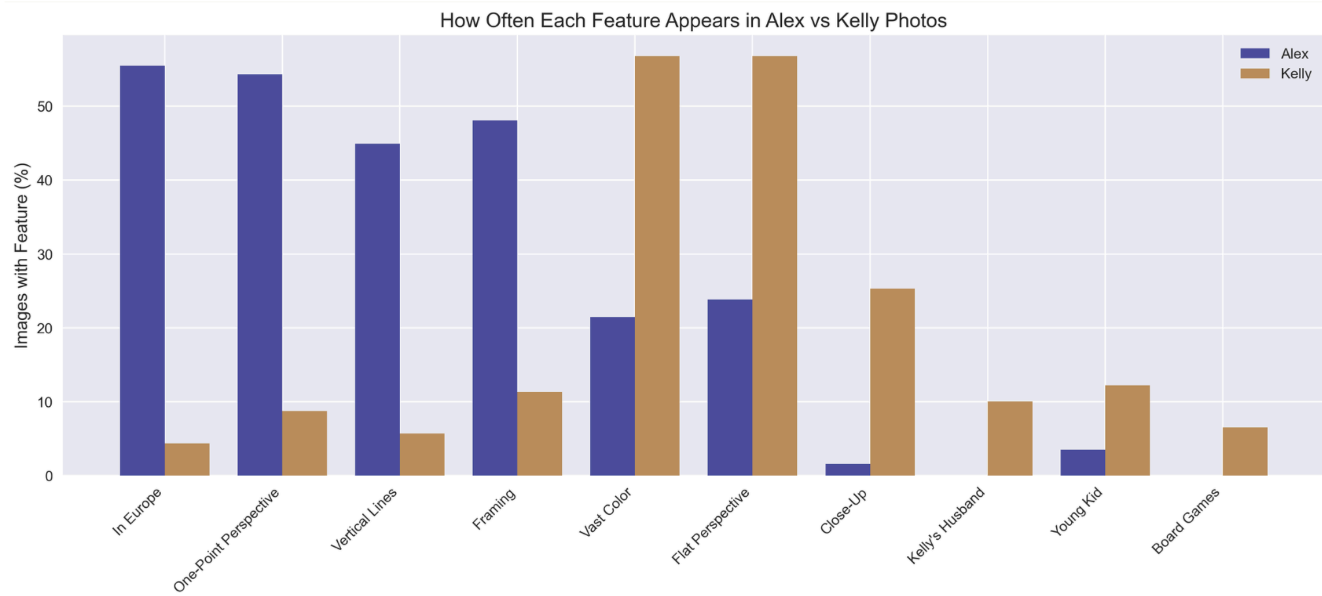


Figure 1. Random Forest Features

To better understand the source of the observed classification behavior and class-level asymmetry between Alex and Kelly's photographs, we also analyzed the manually labeled features using a Random Forest classifier to estimate feature importance. Figure 1 shows the relative frequency of each feature in Alex and Kelly's images across the entire dataset.

Alex's images are strongly associated with composition-based features such as European location, one-point perspective, vertical lines, and framing. In contrast, Kelly's images exhibit a mix of content-based and compositional features, including close-up shots, presence of family members, and wider color distribution. These findings support earlier observations that Alex's photographic style is more compositionally consistent, while Kelly's images display greater visual diversity. The alignment between these feature patterns and model errors suggests that the DINOv2 embeddings capture similar visual cues.

Next Steps + Future Work

A limitation of our approach is that DINOv2 learns general visual representations and does not capture problem specific semantic or social context. Humans use contextual and semantic cues to interpret images and can recognize specific individuals, animals, belongings, and places that are associated with certain people. For example, if a human observer sees a young girl in an image, they can recognize her as Kelly's niece, so they classify the image as "Kelly". DinoV2 embeddings do not capture the prior knowledge and social contexts that humans have.

Humans also interpret images from social and situational contexts. One human contextual cue is understanding the difference between a group photo and a selfie. Group photos are often taken in professional and academic settings. Selfies are often taken in casual and personal settings. Individuals may also have preferences on what type of photo they like to take. Group photos were common in Alex's images while selfies were common in Kelly's images. While these contextual differences were identified through manual feature labeling, they were not incorporated into the embeddings used to train the classifier.

A potential improvement to our model would be to incorporate human contextual features. The process for this includes identifying human contextual features, generating embeddings for those features, and concatenating those embeddings along with the 384 embeddings generated from Dino V2. By combining machine learning representations and human semantic context, the classifier can better distinguish whether an image belongs to Alex or Kelly, potentially improving model performance on classification.

Discussion

Our overall accuracy was 77%, meaning that our model was able to capture some of the meaningful stylistic image differences between Alex and Kelly photos in the Dino v2 embedding space. Interpreting the style of an image is subjective as it involves analyzing multiple image features rather than one visual cue, which is why our accuracy was relatively moderate.

Kelly had a high precision of 0.821 and lower recall of 0.657, which conveys that a good amount of Kelly's photos are incorrectly classified as Alex. This suggests that Kelly's images have greater visual diversity. Greater visual diversity means that there are more visual patterns present in her images, including different compositions, colors, subjects, and perspectives. Alex had a larger recall value of 0.872, suggesting that his photo styles are consistent. On the embedding map, Kelly's embedding points will be spread out while Alex's embedding points will be concentrated in a certain region on the embedding map. If any of Kelly's images have visual components that are similar to Alex's image, then Kelly's embedding point will be close to Alex's region of embedding points. As a result, the SVM will misclassify Kelly's image as Alex, where the decision boundary favors Alex.

According to the feature correlation findings, Alex images are highly correlated with "In Europe" (photos in Europe can have similar compositional features), "One Point Perspective", "Vertical Lines" and "Framing". Kelly has correlations with semantic elements like her child, her husband, board games and the compositional elements. Kelly correlations are mixed with compositional and content related elements, supporting that Kelly has great visual diversity in her images while Alex's images are highly correlated with compositional patterns. Since Alex's images had strong correlations with visual components, Kelly's images with visual components are likely to get misclassified as Alex, which is an overlap in the dino v2 embedding space. This implies that dino v2 was not explicitly trained on human labels but is capturing those visual stylistic features.

Conclusions

The goal of this project was to determine whether a machine learning model could distinguish between two photographers, Alex and Kelly, using only visual information from their images. Using DINOv2 embeddings paired with an SVM classifier, we showed that this task is possible, our model was able to achieve a 77% accuracy on our development set and was able to distinguish separation between Alex and Kelly. This shows that pretrained vision transformers are in fact able to capture differences in photographic style.

Our results suggest that Alex's images are typically more visually similar and are therefore easier for the model to distinguish when compared to Kelly's. We can see this in both our results and confusion matrices, where Kelly's images are more commonly labeled as Alex.

To better understand the differences in Alex and Kelly's photos, we manually labeled images with both content and composition based features. This analysis helped us discover that there certainly were noticeable differences in the photographic styles, and although these features weren't incorporated into our pipeline, they helped give us insight on how to differentiate Alex and Kelly's styles.

Overall, our approach helped illustrate the effectiveness of using a pretrained vision transformer such as DINOv2 as well as its drawbacks. Our pipeline showed that DINOv2 embeddings were able to perform at a reasonable level on a limited amount of data, but it largely missed out on problem specific context. Looking forward, adding in our own information that pertains to our issue may help increase accuracy.

Resources

Alex and Kelly. (2024). Lab 3 Image Dataset [Data set]. Retrieved from https://www.dropbox.com/scl/fo/wfr8hje2hqkxl8jcbh0r0/ACI0i0gaCgEJHFk0MBoXF3w?dl=0&rlkey=4uaibty3m5hcnz1giii023tjm

Gallagher, J. (2023, May 30). *How to classify images with DINOv2*. Roboflow Blog. https://blog.roboflow.com/how-to-classify-images-with-dinov2/

Meta AI. (n.d.). DINOv2 [Web application]. https://dinov2.metademolab.com/

Appendix A - Hand-Picked Features

Composition-based features:

Vertical lines emphasis

Framing

One-point-perspective

Flat perspective

Vast color distribution

Content-based features:

Young children

Kelly’s husband

Board Games

European Look

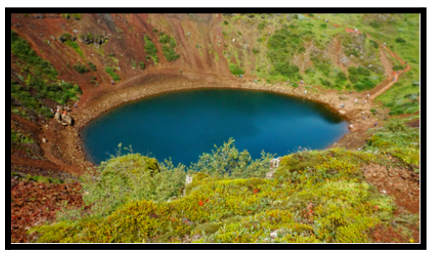
Close-up of people

Appendix B - Error Analysis

Predicted / True	**Alex**	**Kelly**
Alex	**8**	2
Kelly	6	**4**

Table 6. SVM Holdout Set 1 Confusion Matrix

2 images of Alex's misclassified as Kelly's:



6 images of Kelly's misclassified as Alex's:

Appendix C - Model Comparison Experiments

Model	**Training Strategy**	**Holdout 2 Accuracy**
Model 1	Train + CV	0.543
Model 2	Train + Dev	0.565
Model 3	Train + Dev + Holdout 1	0.565